Assignment 9: Machine Learning on AWS

Overview

This assignment offers a comprehensive exploration of both AWS managed AI services and custom deep learning workflows. You will leverage a suite of powerful cloud-based tools, including **Amazon Rekognition** for computer vision, **Amazon Comprehend** for natural language processing**, Amazon Transcribe and Polly** for speech operations, and **Amazon Textract** for document analysis. Beyond managed services, you will gain hands-on experience training and evaluating an AlexNet deep learning model on the MNIST handwritten digit dataset, utilizing Amazon S3 to efficiently store your data and model artifacts for inference. Finally, you will synthesize your work by interpreting the specific outputs of these services, such as labels, sentiments, and transcripts, and summarizing your findings to demonstrate a practical understanding of applied AI on AWS.

Deliverables

Each student (or group) must submit the following:

1. **Presentation Slides**
    - Title slide with student name(s), course, and assignment title.
    - Clear explanation of each completed task:
        - Task 1– Common Setup
        - Task 2 – Part A – Vision
        - Task 3 – Part B – Language with Amazon Comprehend
        - Task 4 – Part C – Speech with Amazon Transcribe & Amazon Poll
        - Task 5 – Part D – Documents with Amazon Textract
        - Task 6 – Reflection Questions
    - Screenshots or videos of key steps/results.
    - Reflection question responses.
    - Lessons learned and challenges encountered.

Presentation Guidelines

- Use professional formatting (legible fonts, consistent colors, organized layout).
- Ensure clarity: avoid overcrowding slides with text; use bullet points and visuals.
- Provide screenshots or videos as evidence of work completed.
- Keep presentations within the time limit (8–11 minutes).

- Be prepared to answer questions about the process and challenges.

1. Task Completion (60 points total)

Each AWS task is evaluated based on correctness, completeness, and evidence (e.g., screenshots/videos output).

2. Presentation Quality (30 points total)

- Clear structure (intro, results, discussion, conclusion)
- Slides visually organized and professional, for example, adding pictures, transitions, animations
- Screenshots or videos included as evidence
- Proper timing (within 8–11 min)

3. Professionalism & Delivery (10 points total)

- Confident and clear presentation delivery
- Ability to answer questions from peers/instructor
- Team collaboration and equal participation

Submission Instructions

- Submit slides (PowerPoint or PDF) to Moodle by the deadline.

Task 1: Common Setup

Before starting the modality-specific parts, complete these steps.

Instructions:

1. Create one S3 bucket to be used across the assignment, e.g.:
    - **Name:** ml-<your_group>
    - **Region:** Same as your SageMaker / AI services.
2. Create the following folders (prefixes) inside the bucket:
    - data/
    - models/
    - outputs/
    - documents/
    - audio/
    - images/

Task 2: Part A – Vision

This part has two components:

1. Using Amazon Rekognition to analyze images.
2. Training AlexNet on the MNIST handwritten digits dataset using PyTorch on SageMaker.

Instructions:

A.1. Vision with Amazon Rekognition

Use **Rekognition** to detect labels and text in images. Rekognition is a cloud-based image and video analysis service that lets you detect objects, people, text, and scenes via simple API calls.

Task A.1.1: Detect Labels via Console

1. Open **Amazon Rekognition** console.
2. Go to **Image → Try demo**.
3. Find 3 images on Internet (e.g., campus scenes, objects, street photos) or choose one of your uploaded images.
4. Run **Detect labels**.
5. For each image, record:
    - The **top 5 labels**.
    - Their **confidence scores**.

Task A.1.2: Detect Text

1. In the same console, run **Detect text** on any image that contains text (signs, posters, etc.).
2. Record:
    - The text detected.
    - Any mis-detections or missing parts.

Task A.1.3: Report

- A table summarizing each image's top labels and confidences.

A.2. Vision: Training AlexNet on MNIST (Deep Learning on SageMaker)

Now you will **train a deep learning model (AlexNet)** to recognize handwritten digits using the **MNIST database of handwritten digits**.

The canonical MNIST dataset (60,000 training, 10,000 test images of 28×28 grayscale digits 0–9) is described and hosted here:

- **MNIST homepage:** https://www.kaggle.com/datasets/hojjatk/mnist-dataset

(You will download it programmatically using PyTorch, but you can browse the page to understand the dataset format.)

Task A.2.1. Launch a SageMaker Notebook

1. Open **Amazon SageMaker Unified Studio**.

2. Use a **Notebook Instance** (e.g., ml.t3.medium or GPU instance if available) with a PyTorch kernel.
3. Make sure the notebook has access to the internet so it can download MNIST.

Optional reading / reference: AWS's own MNIST PyTorch example in SageMaker. Amazon SageMaker Examples

Task A.2.2. AlexNet + MNIST Training Code

Create a new cell and paste the **complete training script:** train.py.

Important: Update **BUCKET_NAME** with your S3 bucket name.

Run the cell and wait for training to complete. Note the **final test accuracy** printed.

Task A.2.3. Using the Model for Inference

Create another cell to define a helper function to predict digits from images. Import the evaluation script: eval.py.

Use the provided images to evaluate the newly created model: alexnet_mnist.pth.

- sample_digit_{number}.png

Task 3: Part B – Language with Amazon Comprehend

Use Amazon Comprehend, an NLP service that detects entities, key phrases, sentiments, and language in text.

You will analyze a small set of text documents (e.g., product reviews, tweets, or comments).

Instructions:

1. Open **Amazon Comprehend** console.
2. Go to **Real-time analysis** (or "Try demo"), and paste a sample of 3–5 lines from your file.
3. Run:
    - **Detect sentiment**.
    - **Detect key phrases**.
    - **Detect entities** (if available).

4. Take screenshots of the output and explain what they are.

Task 4: Part C – Speech with Amazon Transcribe & Amazon Polly

Use **Amazon Transcribe** to convert speech to text, then optionally use **Amazon Polly** to turn text back into speech.

Instructions:

Task C.1. Record Audio & Upload

1. Record a **30-60 second** audio clip reading a paragraph of Vietnamese text.
    - Save as .wav or .mp3.
2. Upload the file to S3 under audio/speech.wav.

Task C.2. Transcribe with Amazon Transcribe Console

1. Open **Amazon Transcribe** console.
2. Create a new **Transcription job**:
    - Job name: ml-<your_group>-job.
    - Language: Vietnamese.
    - Input file: your audio file in S3.
    - Output: either default (Transcribe-managed) or S3 location (audio/output/ in your bucket).
3. Start the job and wait until it is **Completed**.
4. View the transcript and copy the recognized text.

Task C.3. Text-to-Speech with Amazon Polly

1. Open **Amazon Polly** console.
2. Paste a short text (for example, a 2–3 sentence summary of your transcript).
3. Select a voice and click **Listen**.
4. Save the audio file.

Task 5: Part D – Documents with Amazon Textract

Use **Amazon Textract** to extract structured information (text, key-value pairs, tables) from a scanned document (e.g., invoice, receipt or form). Textract goes beyond basic OCR by understanding forms and tables.

Instructions:

1. Open **Amazon Textract** console.
2. Select Analyze Document.
3. Explain the following tabs for a sample documents:
    1. Raw texts
    2. Layout
    3. Tables
    4. Queries
4. **Remember to delete any AWS resoures (e.g. S3, Sagemaker Notebook,…) that you have created to voice incurring costs.**

Task 6: Reflection Questions

Answer the questions below.

Required Question 1

Pick **two** of the AWS AI services you used in this assignment (from: Rekognition, Comprehend, Transcribe, Polly, Textract).

For each chosen service:

1. Describe **one realistic real-world application** (e.g., in healthcare, education, ecommerce, or your university).
2. Explain:
    - What **data** it would process.
    - What **outputs** it would generate.
    - At least **one benefit** and **one challenge** (e.g., accuracy, privacy, latency, cost).

Required Question 2

Consider the **AlexNet MNIST model** you trained in SageMaker and compare it with **using a managed API** (like Rekognition or another AWS AI service) in a hypothetical real application.

- Imagine your team is building a feature that needs to classify images or text for a production system.
- Answer:
    1. When would you prefer to use a **custom model** (like your AlexNet) instead of a managed service?
    2. When would you prefer to use a **managed AI API**?